

Campus Budget & Calorie Meal Planner Summary

Project Purpose

FoodWiseU and George Mason University (GMU) are helping college students balance their food budgets with their daily nutrition goals. College students can have limited time and money, which can make it difficult to make balanced food choices. Students may also consume too many or too few calories during the day.

So the campus budget and calorie meal planner is a Java application designed to help students make budget friendly dining decisions. The program tracks both the amount of money remaining in students' food budget and their total daily calorie intake. So the user enters their daily meal budget and then selects their gender for the calorie goal, afterwards chooses meals throughout the day. The program then calculates the cost and calories for each single meal and provides the user with updated information.

Key Design Decisions

So one of our key design decisions was using JOptionPane. JOptionPane really helps provide a simple visual way for users to enter information and make meal recommendations and selections. So if you see this really supports a project goal and really allows students to quickly enter their information and track their meals without wasting any valuable time.

So if we look deeper into this, another design decision was using fixed constants for breakfast, dinner, and lunch. The program is really set up with meal costs and calorie values for the available meals. Since our students said that menus can change daily, but they often continue choosing their regular meals, having set meal prices and calorie values makes it easier for the program to quickly calculate the information for the meal selection.

Therefore, the program also uses variables to track the remaining budget and the total calories. The remaining budget starts with the user's entered budget, while the total calories starts at zero. When the meal is selected, its cost is subtracted from the remaining budget and the calories are added to the daily calorie counter.